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Aggregation-based operations for reversal fuzzy switch graphs

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Abstract

Fuzzy Switch Graphs (FSGs) are reactive fuzzy graphs that model systems in which accessibility relations and fuzzy values are changed whenever an edge is crossed [19]. Reversal Fuzzy Switch Graphs ($RFSG$s) were presented in [6] and model fuzzy reactive systems which provide the activation and deactivation of resources, a functionality that FSGs do not offer [19]. Activation and deactivation of arrows in a switch graph makes its applicability wider in fields like computer science and engineering. In this sense, the definition of operations for $RFSG$s is an important issue, since it enables to understand the algebraic structure of these graphs and allows to generate new model of systems from the previous one. This paper introduces some aggregation-based operations: union, intersection and extension for $RFSG$s. For each operation, the paper verify some properties. It ends with an application for engineering.

Keywords: Fuzzy systems; Fuzzy switch graphs; Reversal fuzzy switch graphs; Reactive systems; Aggregations functions

1. Introduction

A graph is a representation of a binary relation between objects. Given a graph, if its nodes (worlds) or edges (accessibility relations) are changed whenever a movement is made between its nodes, then we will say that it is a reactive graph. This concept has been introduced by several authors such as Benthen [21], Areces [1] and Gabbay [9][11].

In order to model reactivity, Gabbay et al. in [10] presented graphs with the addition of a new type of edges, called high-order arrows. These edges relate usual edges (nodes to nodes) to usual edges and usual edges to high-order arrows in order to modify the characteristics of the target edge whenever its source edge is crossed. Graphs with high-order arrows in their structure were called *Switch Graphs*.

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In the real world, there are state-based systems with uncertain behavior (fuzziness) and such that the transition from one state to another entails a system reconfiguration. Biological systems, fluid dynamics systems are some examples. In 2020, Santiago et al. [19] extended the notion of *Switch Graphs* to *Fuzzy Switch Graphs* (FSGs). FSGs use aggregation functions to promote the system update. For systems which require different aggregations for different edges, Santiago et al. introduced the *Fuzzy Reactive Graphs* (FRGs).

FSGs and FRGs, however, cannot model systems in which a movement activate or deactivates a transition between states (see the case of the Tweetie in [10]). In this context, in 2020, Campos et al. [6] introduced this resource to FSGs and FRGs, giving rise to *Reversal Fuzzy Switch Graphs* ($RFSG$s). FSGs and $RFSG$s have an associated fuzzy graph, called induced fuzzy graph, which are the connection with fuzzy graphs theory [19], [17].

This paper introduces some operations for $RFSG$s where we stand out the extension, which enables us to model the extension of a system to a move ampler one. The operations take into account the effect of the corresponding aggregation in each updating. So an investigation of the effect of aggregations in each movement is shown.

The paper is organized in the following way: section 2 presents some basic concepts about aggregations functions. Section 3 present the notion of $RFSG$s and how its reactivity is modeled through the use of aggregations. Section 4 introduces the operations of union and intersection based on aggregations, and the operation of extension of $RFSG$s. Section 5 shows how the extension of $RFSG$s can be used to model a coupled of two dynamic control systems. Finally, section 6 provides some final remarks.

2. Aggregation functions

In this section we present the notion of aggregation functions, their main properties and some examples. For a more detailed information see [2], [4], [7], [12] and [15]. Some applications of aggregations can be seen at [5], [14], [18] and [20].

Definition 1 *(Aggregation functions).* A n-ary function $A : [0,1]^n \to [0,1]$ is an **aggregation function** (or just **aggregation**) whenever:

i) $A(0,0,...,0) = 0$ and $A(1,1,...,1) = 1$,
ii) $x_i \leq y_i$, for $i = 1,...,n$, implies $A(x_1,x_2,...,x_n) \leq A(y_1,y_2,...,y_n)$ for all $(x_1,x_2,...,x_n),(y_1,y_2,...,y_n) \in [0,1]^n$ (isotonicity).

Example 2. The functions: $A_n(\bar{x}) = \frac{1}{n}(x_1 + ... + x_n)$ (*arithmetic mean*), $A_n(\bar{x}) = x_1 \cdot ... \cdot x_n$ (*product*), $A_n(\bar{x}) = \sqrt[n]{x_1 \cdot ... \cdot x_n}$ (*geometric mean*), $A_n(\bar{x}) = max(x_1,x_2,...,x_n)$ (maximum), $A_n(\bar{x}) = min(x_1,x_2,...,x_n)$ (minimum) and projections, $\Pi_j : [0,1]^n \longrightarrow [0,1]$, s.t. $\Pi_j(x_1,...,x_j,...,x_n) = x_j$ are aggregation functions.

Definition 3. A n-ary aggregation A is **idempotent** if $A(x,x,...,x) = x$, for each $x \in [0,1]$. If $n = 2$ (A is binary), A is **associative** whenever $A(A(x_1,x_2),x_3) = A(x_1,A(x_2,x_3))$ holds for all x_1,x_2,x_3 in $[0,1]$ and A is **commutative** whenever $A(x_1,x_2) = A(x_2,x_1)$ holds for all x_1,x_2 in $[0,1]$.

Example 4. The *arithmetic mean*, *geometric mean*, *minimum* and *maximum* are idempotent aggregations. The *product*, *minimum* and *maximum* are associative aggregations.

Definition 5. An aggregation A has a **neutral element** $e \in [0,1]$ if, for every $x \in [0,1]$, $A(e,...,e,x,e,...,e) = x$.

Example 6. The *minimum* has neutral element $e = 1$ and the *maximum* has neutral element $e = 0$.

Proposition 7. *[4] Consider three aggregations:* $A_1, A_2 : [0,1]^n \to [0,1]$ *and* $\nabla : [0,1]^2 \to [0,1]$. *The application* $\nabla(A_1,A_2) : [0,1]^n \to [0,1]$ *s.t.*

$$\nabla(A_1,A_2)(\bar{x}) = \nabla(A_1(\bar{x}),A_2(\bar{x}))$$

for $\bar{x} \in [0,1]^n$ *is also an aggregation function.*

3. Reversal fuzzy switch graphs and reactivity

In 2020, Santiago et al. [19] introduced the notion of *Fuzzy Switch Graphs* (FSGs) and used then to model some biological applications (*Circadian rhythm in cyanobacteria* and autoregulative protein). This work applied aggregation functions in order to combine fuzzy values of graph edges. The FSGs, however, are not able to model reactive systems which requires the activation and deactivation of transitions. In order to overcome that, this section presents the notion of *Reversal Fuzzy Switch Graphs* ($RFSG$s) [6].

Definition 8 *(Fuzzy graphs [13]).* A **fuzzy graph** is a structure $\langle V, R\rangle$, such that V is a non-empty set of vertices and R is a fuzzy set $R : V \times V \to [0, 1]$.

Remark 1. In [13], [16] the possibility of a membership function $\upsilon : V \to [0, 1]$ assigning fuzzy values to vertices of the graph is mentioned. However, [13] defines fuzzy graphs assuming the set of vertices V as a crisp set. In this paper, we also assume this definition for the different types of fuzzy graphs presented here. The Fig. 1(a) shows a fuzzy graph.

In 2004, Gabbay et al. [9] introduced the idea of an enriched graph. Gabbay et al. use double arrows to express the connection between arrows. The double arrows deactivate its target arrows. In 2012, Gabbay and Marcelino [10] extended the work of Gabbay by introducing *Switch Graphs*.

Definition 9 *(Switch graphs [3] [9]).* A **switch graph** is a pair $\langle W, R\rangle$ s.t. W is a non-empty set (set of **worlds**) and $R \subseteq A(W)$ is a set of arrows, called **switches**, where $A(W) = \bigcup_{i\in\mathbb{N}} A_i(W)$ with

$$\begin{cases} A_0(W) = W \times W, \\ A_{i+1}(W) = A_0(W) \times A_i(W). \end{cases} \tag{1}$$

In [19], Santiago et al. provided the fuzzification for switch graphs with some modifications.

Definition 10 *(Fuzzy switch graphs [19]).* Let W be a non-empty finite set (set of **states** or **worlds**) and the set $S = \bigcup_{n\in\mathbb{N}} S^n$ where $S^0 \neq \emptyset$ and,

$$\begin{cases} S^0 \subseteq W \times W, \\ S^{n+1} \subseteq S^0 \times S^n. \end{cases} \tag{2}$$

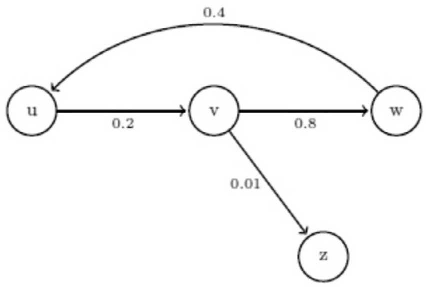
A **fuzzy switch graph (FSG)** is a pair $\mathcal{M} = \langle W, \varphi : S \to [0, 1]\rangle$, s.t. φ is a membership function for S. The elements $a_i^0 \in S^0$ ($i \in \mathbb{N}$) are called **zero-order arrows**. The elements of S^{n+1}, $n \geq 0$, are called **high-order arrows**.

The Fig. 1(b) shows a *Fuzzy Switch Graph* $\mathcal{M}$ with $S^0 = \{(u, v), (v, w), (w, u), (v, z)\}$, $S^1 = \big\{\big((v, w), (w, u)\big), \big((v, z), (v, w)\big)\big\}$ and $S^2 = \big\{\big((u, v), \big((v, w), (w, u)\big)\big)\big\}$. Associated to S^0, S^1 and S^2, we have the following membership values: $\varphi(u, v) = 0.2$, $\varphi(v, w) = 0.8$, $\varphi(v, z) = 0.01$, $\varphi(w, u) = 0.4$, $\varphi\big((v, w), (w, u)\big) = 0.7$, $\varphi\big((v, z), (v, w)\big) = 0.1$, $\varphi\big((u, v), \big((v, w), (w, u)\big)\big) = 0.1$ and $\varphi(a) = 0$, for all $a \in S^j$ and $j > 2$. The reactivity of the FSGs is modeled in such a way that: *whenever a zero-order arrow is crossed, the fuzzy values of the graph are updated.*

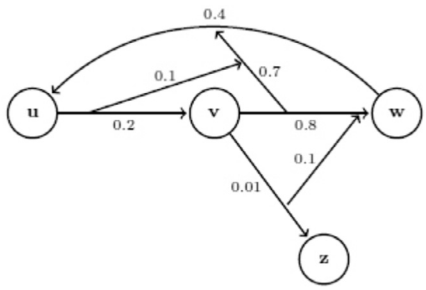
The graph is updated in the following way: *given a high-order arrow h with a source arrow s and a target arrow t, the new value of t, after s has been crossed, is the aggregation of its value with the values of h and s.*

Definition 11. [19] Given an aggregation function $A : [0, 1]^3 \to [0, 1]$ and a FSG $\mathcal{M} = \langle W, \varphi : S \to [0, 1]\rangle$, the structure $\mathcal{M}_A^{a_i^0} = \langle W, \varphi_{a_i^0}^A : S \to [0, 1]\rangle$ s.t.

$$\varphi_{a_i^0}^A(a) = \begin{cases} \varphi(a), & \text{if } (a_i^0, a) \notin S \\ A\left(\varphi(a_i^0), \varphi(a_i^0, a), \varphi(a)\right), & \text{otherwise} \end{cases} \tag{3}$$



(a)



(b) $\mathcal{M}$

Fig. 1. (a) Fuzzy Graph and (b) Fuzzy Switch Graph $\mathcal{M}$.

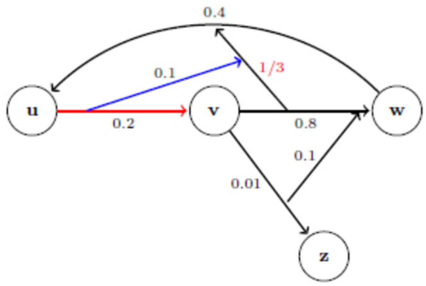


Fig. 2. Reconfiguration of Fig. 1(b) after crossing (u,v) i.e. $\mathcal{M}^{(u,v)}_{arith}$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

is called the **reconfiguration of $\mathcal{M}$, based on A, after crossing a^0_i**.

Remark 2. Any reconfiguration of a FSG $\mathcal{M}$ is a FSG.

Example 12. Fig. 2 shows the reconfiguration, $\mathcal{M}^{(u,v)}_{arith}$, of the Fig. 1(b) obtained after crossing (u,v) by using the arithmetic mean. In this case, $A(0.2, 0.1, 0.7) = (0.2 + 0.1 + 0.7)/3 = 1/3$.

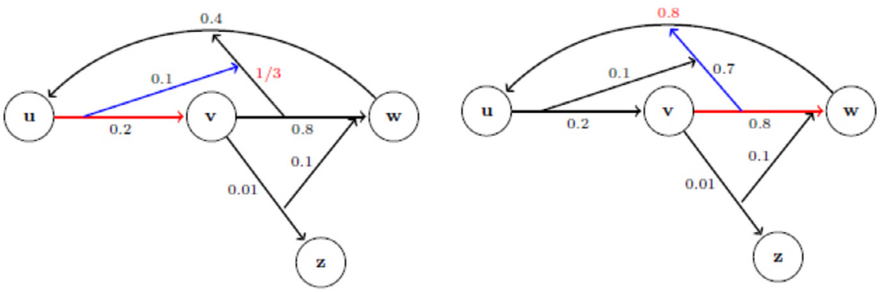
A FSG is endowed with only one aggregation, however Santiago et al. [19] extended this feature in order to enable multiple aggregation.

Definition 13 *(Fuzzy reactive graphs [19]).* Let be a FSG $\mathcal{M} = \langle W, \varphi : S \to [0,1]\rangle$, a set of aggregations $A = \{A_1, ..., A_k : [0,1]^3 \to [0,1]\}$ and a function $Ag : S_\to \to A$ for $S_\to = \{a^0_i \in S^0 ; (a^0_i, a) \in S,\ \text{with } a \in S\}$. The pair $\mathcal{M}_R = \langle \mathcal{M}, Ag\rangle$ is called **fuzzy reactive graph (FRG)**. Given $a^0_i \in S^0$, **the reconfiguration of $\mathcal{M}_R$ after crossing** a^0_i is the FRG $\mathcal{M}^{a^0_i}_R = \langle \mathcal{M}^{a^0_i}_{Ag}, Ag\rangle$ s.t.

$$\varphi^{Ag}_{a^0_i}(a) = \begin{cases} Ag(a^0_i)\big(\varphi(a^0_i), \varphi(a^0_i, a), \varphi(a)\big), & \text{if } (a^0_i, a) \in S; \\ \varphi(a), & \text{otherwise.} \end{cases} \tag{4}$$

Example 14. Let be the FSG in Fig. 1(b), consider $S^0 = \{a^0_1 = (u,v),\ a^0_2 = (v,w),\ a^0_3 = (v,z),\ a^0_4 = (w,u)\}$ and $A = \{arith,\ max\}$. Making $Ag(a^0_1) = arith$ and $Ag(a^0_2) = Ag(a^0_3) = max$, the structure $\mathcal{M}_R$ is a FRG. Fig. 3 contains $\mathcal{M}^{a^0_1}_R$ and $\mathcal{M}^{a^0_2}_R$, i.e. the reconfiguration of $\mathcal{M}_R$ after crossing a^0_1 and a^0_2, respectively.

Gabbay et al. [10] classify the high-order arrows of *switch graphs* in two types: connecting arrows and disconnecting arrows. As the names suggest, when the source arrow of a connecting/disconnecting arrow is crossed, its target



(a) $\mathcal{M}_R^{a_1^0}$ (b) $\mathcal{M}_R^{a_2^0}$

Fig. 3. Reconfigurations of $\mathcal{M}_R$.

arrow is activated/deactivated. In this context, we realize that the high-order arrows in FSGs and FRGs do not have this feature. In [6], the first steps were taken in order to endow FSGs and FRGs with this kind of high-order arrows.

Definition 15 *(Reversal fuzzy switch graphs [6]).* Let W be a non-empty finite set of **states** or **worlds**, the following family of sets defined recursively:

$$\begin{cases} S^0 \subseteq W \times W, \\ S^{n+1} \subseteq S^0 \times S^n \times \{\bullet, \circ\}. \end{cases} \tag{5}$$

s.t. $S^0 \neq \emptyset$ and for any $n \geq 1$, $(a_i^0, a, \bullet) \notin S^n$ or $(a_i^0, a, \circ) \notin S^n$. Consider $S = \bigcup_{n \in \mathbb{N}} S^n$, a **reversal fuzzy switch graph (RFSG) is a pair** $M = \langle W, \mu : S \to [0,1] \times \{\text{ON}, \text{OFF}\} \rangle$.[1] Arrows with $\bullet$ in the third component are called **connecting** arrows and arrows with $\circ$ in the third component are called **disconnecting** arrows.

In [10] the case in which a connecting and disconnecting arrow act simultaneously over the same arrow is avoided by choosing one of the two types of high-order arrows as the prevalent arrow, however Definition 15 follows another way. It states that there won't be a connecting and disconnecting arrow, with the same source arrow, acting over the same target arrow in a $RFSG$.

Pictorially, active arrows have a normal line whereas inactive arrows have a dashed line. Connecting (disconnecting) arrows change the targeted arrow state for active (inactive) and are drawn with a black (white) arrowhead.

Example 16. Fig. 4 shows a $RFSG$.

In order to ensure readability, we introduce some notation and nomenclatures:

- Arrows in S^n will be denoted by a_i^n, for $n \geq 0$ and $i \in \mathbb{N}$.
- In the following, we make an abuse of notation. When necessary and if the context is clear, we will denote in more detail the arrows in S^n in a more expanded way. For example, a_i^0 from x to y will be denoted by $[xy]$, the disconnecting and connecting first-order arrows from $[xy]$ to $[uv]$ will be denoted by $[\![[xy],[uv],\circ]\!]$ and $[\![[xy],[uv],\bullet]\!]$, respectively. When referring to any high-order arrow, we write $\sigma \in \{\circ, \bullet\}$ instead of $\circ$ or $\bullet$. For example, any first-order arrow from $[uv]$ to $[xy]$ will be written $[\![[uv],[xy],\sigma]\!]$. Any second-order arrows from $[zw]$ to $[\![[xy],[uv],\sigma]\!]$ will be denoted by $[\![[zw],[\![[xy],[uv],\sigma]\!],\sigma']\!]$ $(\sigma' \in \{\circ, \bullet\})$.
- When there is no need to specify the order of the arrow belonging to set S, we will denote $a \in S$.
- Given the projection functions $\Pi_1 : [0,1] \times \{\text{ON}, \text{OFF}\} \to [0,1]$ and $\Pi_2 : [0,1] \times \{\text{ON}, \text{OFF}\} \to \{\text{ON}, \text{OFF}\}$, if $a \in S$ we write $\mu_1(a) = \Pi_1(\mu(a))$ and $\mu_2(a) = \Pi_2(\mu(a))$ to indicate the first and second components of $\mu(a)$.

[1] In this paper we assume that the membership function is valued in the complete lattice $[0.1] \times \{\text{ON}, \text{OFF}\}$ using the product order where $\text{OFF} \leq \text{ON}$.

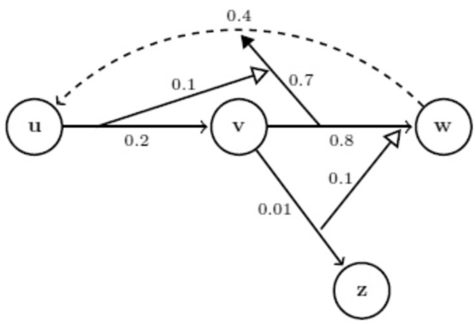


Fig. 4. $RFSG\ M$.

- Let $R \subseteq S$, the set of active arrows in R is denoted by $R^*_\mu := \{a \in R;\ \mu_2(a) = \text{ON}\}$ (When it's clear in context, we will just write R^*) and the set of source arrows in R is denoted by

$$R_{\rightarrow} = \left\{a_i^0 \in R;\ [\![a_i^0, b, \sigma]\!] \in S \text{ with } b \in S \text{ and } \sigma \in \{\circ, \bullet\}\right\}.$$

Like FSGs, $RFSG$s make use of aggregations to model the reactivity of systems, however for $RFSG$s the updating process takes into account the action of connecting and disconnecting arrows. The general idea is: *whenever an active zero-order arrow is crossed, the fuzzy values and the arrows type (active or inactive) of a target arrow is updated.*

In the following, we fix a $RFSG\ M = \langle W, \mu : S \rightarrow [0,1] \times \{\text{ON}, \text{OFF}\}\rangle$.

Definition 17. Given M and an aggregation function $A : [0,1]^3 \rightarrow [0,1]$, a $RFSG$ **based on** A **after crossing an active zero-order arrow** a_i^0, is the structure $M^A_{a_i^0} = \langle W, \mu^A_{a_i^0} : S \rightarrow [0,1] \times \{\text{ON}, \text{OFF}\}\rangle$ s.t.

$$\mu^A_{a_i^0}(a) = \begin{cases} \Big(A\big(\mu_1(a_i^0), \mu_1([\![a_i^0, a, \bullet]\!]), \mu_1(a)\big), \text{ON}\Big), & \text{if } [\![a_i^0, a, \bullet]\!] \in S^* \\ \Big(A\big(\mu_1(a_i^0), \mu_1([\![a_i^0, a, \circ]\!]), \mu_1(a)\big), \text{OFF}\Big), & \text{if } [\![a_i^0, a, \circ]\!] \in S^* \\ \mu(a), & \text{otherwise} \end{cases} \tag{6}$$

$M^A_{a_i^0}$ is called **reconfiguration of** M**, based on** A**, after crossing** a_i^0.

Remark 3. The reconfiguration of a $RFSG\ M$ is a $RFSG$. Furthermore, the arrows contained in S can receive a null fuzzy value. Graphically, these arrows will be displayed only if there is the possibility of modifying this value by some high-order arrow.

Notation 1. In order to make the presentation of the graphs and the movements more intuitive, we establish the following: Arrows that are crossed over the graph will be drawn in red. High-order arrows that act on the graph configuration, after an arrow has been crossed, will be drawn in blue. The first arrow crossed will have a single arrowhead, the second arrow crossed will have a double arrowhead, the third arrow to be crossed will have a triple arrowhead and so on. If multiple movements are made repeatedly on the same arrow, the arrow arrowheads will show the order of the last movement. For example, if the movement is made three times on the same arrow, graphically we will see only a red triple-headed arrow in the graph.

Example 18. Fig. 5 shows the reconfiguration, $M^{arith}_{[uv]}$, of $RFSG\ M$ in Fig. 4.

Note that the previously active arrow $[\![[vw],[wu],\bullet]\!]$ becomes inactive due the action of the disconnection arrow $\big[\!\big[[uv], [\![[vw],[wu],\bullet]\!], \circ\big]\!\big]$.

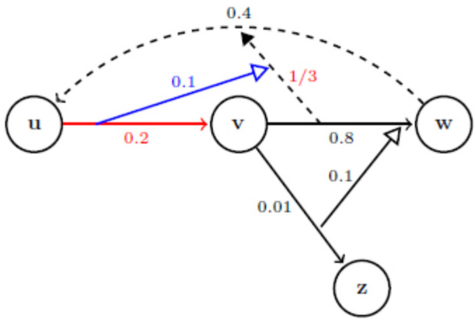


Fig. 5. Reconfiguration $M^{arith}_{[uv]}$ of $RFSG\ M$ (Fig. 4).

Definition 19 *(Reversal fuzzy reactive graphs).* Given M, the set of ternary aggregation functions $A = \{A_1, ..., A_k : [0,1]^3 \to [0,1]\}$ and a function $Ag : S_{\to} \to A$. The pair $M_R = \langle M, Ag\rangle$ is called **reversal fuzzy reactive graph (RFRG)**.

If $a_i^0 \in S^0$ is an active arrow, the **reconfiguration of M_R after crossing** a_i^0 is the $RFRG$ $M_R^{a_i^0} = \langle M^{a_i^0}, Ag\rangle$, where $M^{a_i^0} = \langle W, \mu^{Ag}_{a_i^0}\rangle$ is the $RFRG$ s.t.

$$\mu^{Ag}_{a_i^0}(a) = \begin{cases} \Big(Ag(a_i^0)\big(\mu_1(a_i^0), \mu_1([\![a_i^0, a, \bullet]\!]), \mu_1(b)\big), \text{ON}\Big), & \text{if } [\![a_i^0, a, \bullet]\!] \in S^* \\ \Big(Ag(a_i^0)\big(\mu_1(a_i^0), \mu_1([\![a_i^0, a, \circ]\!]), \mu_1(a)\big), \text{OFF}\Big), & \text{if } [\![a_i^0, a, \circ]\!] \in S^* \\ \mu(a), & \text{otherwise} \end{cases} \tag{7}$$

Remark 4. $RFRG$s generalizes $RFSG$s as they allow relating to different zero-order arrows, source of high-order arrows, different aggregations.

Example 20. Let be the $RFSG\ M$ in Fig. 6(a), we have $S^0 = \{a_1^0 = [uv],\ a_2^0 = [vw],\ a_3^0 = [vz],\ a_4^0 = [wu]\}$ and consider $A = \{arith,\ max\}$. Making $Ag(a_1^0) = arith$ and $Ag(a_2^0) = Ag(a_3^0) = max$, the Fig. 6(b) contains the result of crossing a_i^0 and then a_2^0, which we denote by $M_R^{a_1^0 a_2^0}$. In addition, the Fig. 6(c) shows the reconfiguration of M after crossing just the arrow a_2^0 i.e., $M_R^{a_2^0}$.

4. Operations on $RFSG$s

In what follow we present some operations in order to build a reactive system by combining two pre-existing reactive systems. The operations that follow will be built from arbitrary binary or ternary aggregations.

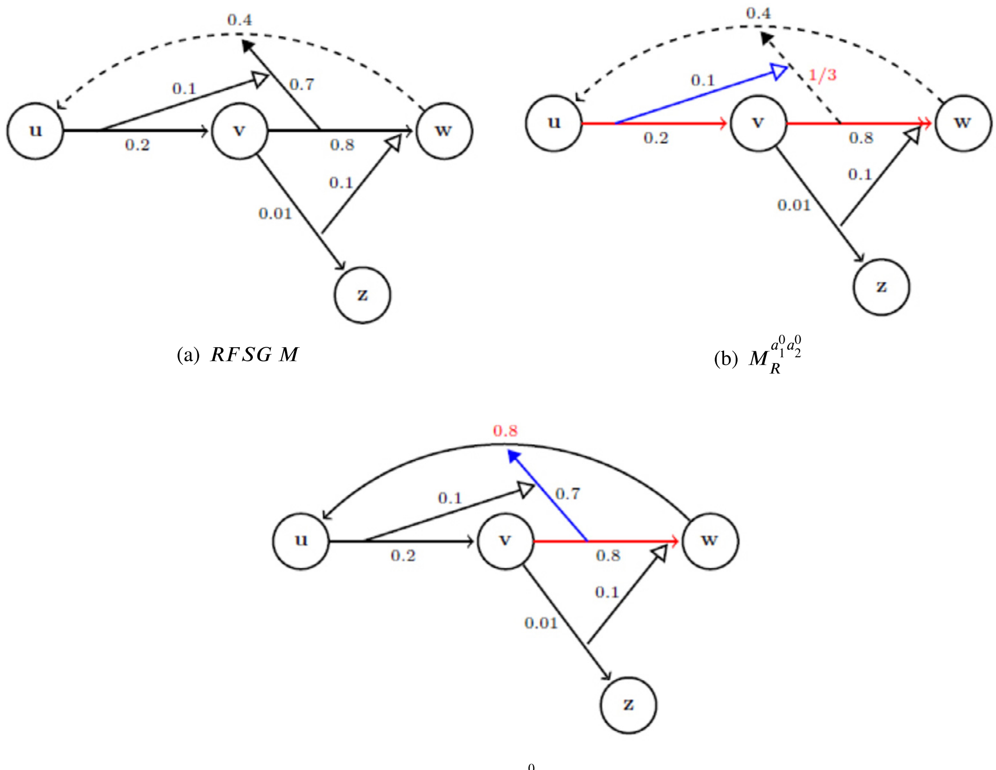
In the following we consider:

- the $RFSG$s $M = \langle W, \mu : S \to [0,1] \times \{\text{ON}, \text{OFF}\}\rangle$ and $N = \langle V, \eta : T \to [0,1] \times \{\text{ON}, \text{OFF}\}\rangle$;
- the $RFRG$s $M_R = \langle M, Ag_M\rangle$ and $N_R = \langle N, Ag_N\rangle$,

s.t., $A_M = \{A_{(1,M)}, A_{(2,M)}, ..., A_{(m,M)} : [0,1]^3 \to [0,1]\}$, $A_N = \{A_{(1,N)}, A_{(2,N)}, ..., A_{(n,N)} : [0,1]^3 \to [0,1]\}$ are sets of aggregation functions and $Ag_M : S_{\to} \to A_M$, $Ag_N : T_{\to} \to A_N$ are functions.

Given two membership functions $\mathcal{X} : X \to [0,1] \times \{\text{ON}, \text{OFF}\}$ and $\mathcal{Y} : Y \to [0,1] \times \{\text{ON}, \text{OFF}\}$, we consider the sets:

- $(\mathcal{X} \cap \mathcal{Y})_{\text{ON}} = \{a \in X \cap Y,\ \mathcal{X}_2(a) = \mathcal{Y}_2(a) = \text{ON}\}$,
- $(\mathcal{X} \cap \mathcal{Y})_{\text{OFF}} = \{a \in X \cap Y,\ \mathcal{X}_2(a) = \mathcal{Y}_2(a) = \text{OFF}\}$,
- $(\mathcal{X} \cap \mathcal{Y})_{\neq} = \{a \in X \cap Y,\ \mathcal{X}_2(a) \neq \mathcal{Y}_2(a)\}$,



(a) $RFSG\ M$

(b) $M_R^{a_1^0 a_2^0}$

(c) $M_R^{a_2^0}$

Fig. 6. $RFSG\ M$ and its reconfigurations.

For abuse of language, the set $(\mathcal{X} \cap \mathcal{Y})_{\text{ON}}$ will be written as $(X \cap Y)_{\text{ON}}$ and the other sets will follow the same pattern.

4.1. Δ-union and Δ-intersection of $RFSGs$

Given an aggregation $\Delta : [0,1]^2 \longrightarrow [0,1]$, let be the following operations:

Definition 21. The structure $M \cup_\Delta N = \langle W \cup V, (\mu \cup_\Delta \eta) : S \cup T \longrightarrow [0,1] \times \{\text{ON}, \text{OFF}\}\rangle$ s.t.

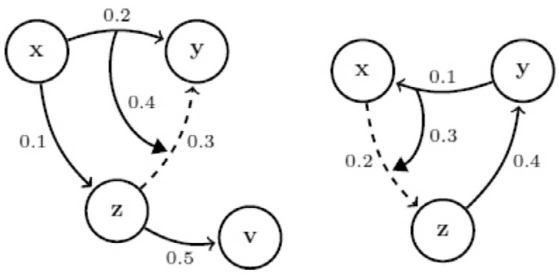
$$(\mu \cup_\Delta \eta)(a) = \begin{cases} \mu(a), \text{ case } a \in S - T \\ \eta(a), \text{ case } a \in T - S \\ \Big(\Delta\big(\mu_1(a), \eta_1(a)\big), \text{ON}\Big), \ \textit{case } a \in (S \cap T)_{\text{ON}} \cup (S \cap T)_{\neq} \\ \Big(\Delta\big(\mu_1(a), \eta_1(a)\big), \text{OFF}\Big), \ \textit{case } a \in (S \cap T)_{\text{OFF}} \end{cases}$$

with M and N $RFSG$s, s.t., for any $n \geq 1$, $[\![a_i^0, a, \bullet]\!] \notin S^n \cup T^n$ or $[\![a_i^0, a, \circ]\!] \notin S^n \cup T^n$, is called **$\Delta$-union** of M and N.

Definition 22. The structure $M \cap_\Delta N = \langle W \cap V, (\mu \cap_\Delta \eta) : S \cap T \longrightarrow [0,1] \times \{\text{ON}, \text{OFF}\}\rangle$ s.t.

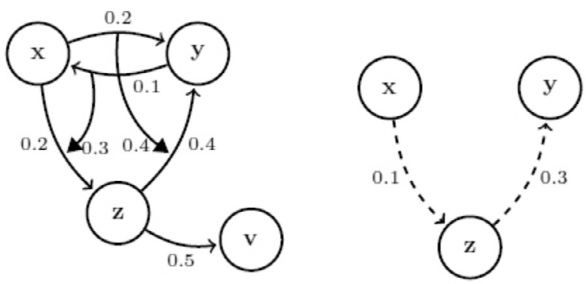
$$(\mu \cap_\Delta \eta)(a) = \begin{cases} \Big(\Delta\big(\mu_1(a), \eta_1(a)\big), \text{ON}\Big), \ \textit{case } a \in (S \cap T)_{\text{ON}} \\ \Big(\Delta\big(\mu_1(a), \eta_1(a)\big), \text{OFF}\Big), \ \textit{case } a \in (S \cap T)_{\text{OFF}} \cup (S \cap T)_{\neq} \end{cases}$$

with M and N $RFSG$s is called **Δ-intersection** of M and N.



(a) M (b) N

Fig. 7. $RSFG$s M and N.



(a) $M \cup_{max} N$ (b) $M \cap_{min} N$

Fig. 8. *max*-union and *min*-intersection of M and N.

Remark 5. The Δ-union and Δ-intersection are $RFSG$s. Furthermore, the Definition 21 only considers $RFSG$s M and N whose Δ-union does not produce connection and disconnection arrows with the same source arrow and target arrow. This restriction guarantees that, the union will satisfy the Definition 15. The same requirement does not occur in Definition 22, since, in order to $[\![a_i^0, a, \circ]\!]$ and $[\![a_i^0, a, \bullet]\!]$ be in the Δ-intersection, both must be in M and N, meaning that they are not $RFSG$s. It is not possible to have the case $W \cap V = \emptyset$.

The above operations combine and establish the following criteria for arrows that are in the $(S \cap T)_{\neq}$: *For Δ-union, we establish the prevalence of the status* ON *for the arrows are active in one graph and inactive in another graph. For Δ-intersections, we establish the prevalence of the status* OFF *for the arrows that are active in one graph and inactive in the other.*

Example 23. Fig. 7 shows two $RFSG$s M and N, whereas the Fig. 8 shows $M \cup_{max} N$ and $M \cap_{min} N$.

In what follows we relate some properties of aggregations with properties of Δ-union and Δ-intersection.

Proposition 24. *Given three $RFSG$s, M, N and $P = \langle K, \gamma : Q \to [0,1] \times \{\text{ON}, \text{OFF}\}\rangle$. The following properties are satisfied:*

i) *$M \cup_\Delta N = N \cup_\Delta M$ and $M \cap_\Delta N = N \cap_\Delta M$ if Δ is commutative;*
ii) *$M \cup_\Delta (N \cup_\Delta P) = (M \cup_\Delta N) \cup_\Delta P$ and $M \cap_\Delta (N \cap_\Delta P) = (M \cap_\Delta N) \cap_\Delta P$ if Δ is commutative and associative;*
iii) *$M \cup_\Delta M = M$ and $M \cap_\Delta M = M$ if Δ is idempotent.*

Proof. Indeed,

i) This proof is straightforward.
ii) $M \cup_\Delta (N \cup_\Delta P) = \langle W \cup (V \cup K), [\mu \cup_\Delta (\eta \cup_\Delta \gamma)] : S \cup (T \cup Q) \to [0,1] \times \{\text{ON}, \text{OFF}\}\rangle$. We know that $W \cup (V \cup K) = (W \cup V) \cup K$. So, we just need show that $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)] = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma]$. Indeed,

Case $a \in S - (T \cup Q)$, $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \mu(a) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$;
Case $a \in T - (S \cup Q)$, $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \eta(a) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$;
Case $a \in Q - (S \cup T)$, $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \gamma(a) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$;
Case $a \in [(S \cap T) - Q]$,

$$\begin{aligned}[\mu \cup_\Delta (\eta \cup_\Delta \gamma)]_1(a) &= \Delta\big(\mu_1(a), (\eta \cup_\Delta \gamma)_1(a)\big)\\ &= \Delta\big(\mu_1(a), \eta_1(a)\big)\\ &= (\mu \cap_\Delta \eta)_1(a)\\ &= [(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a)\end{aligned}$$

Then $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \big([(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a), \text{ON}\big) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$ if $a \in [(S \cap T) - Q]_{\text{ON}} \cup [(S \cap T) - Q]_{\neq}$ or, $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \big([(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a), \text{OFF}\big) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$ if $a \in [(S \cap T) - Q]_{\text{OFF}}$.
Case $a \in [(S \cap Q) - T]$,

$$\begin{aligned}[\mu \cup_\Delta (\eta \cup_\Delta \gamma)]_1(a) &= \Delta\big(\mu_1(a), (\eta \cup_\Delta \gamma)_1(a)\big)\\ &= \Delta\big(\mu_1(a), \gamma_1(a)\big)\\ &= \Delta\big((\mu \cup_\Delta \eta)_1(a), \gamma_1(a)\big)\\ &= [(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a)\end{aligned}$$

Then $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \big([(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a), \text{ON}\big) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$ if $a \in [(S \cap Q) - T]_{\text{ON}} \cup [(S \cap Q) - T]_{\neq}$ or, $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \big([(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a), \text{OFF}\big) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$ if $[(S \cap Q) - T]_{\text{OFF}}$.
Case $a \in [(T \cap Q) - S]$,

$$\begin{aligned}[\mu \cup_\Delta (\eta \cup_\Delta \gamma)]_1(a) &= \Delta\big(\mu_1(a), (\eta \cup_\Delta \gamma)_1(a)\big)\\ &= (\eta \cup_\Delta \gamma)_1(a)\\ &= \Delta\big(\eta_1(a), \gamma_1(a)\big)\\ &= \Delta\big((\mu \cup_\Delta \eta)_1(a), \gamma_1(a)\big)\\ &= [(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a)\end{aligned}$$

Then $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \big([(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a), \text{ON}\big) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$ if $a \in [(T \cap Q) - S]_{\text{ON}} \cup [(T \cap Q) - S]_{\neq}$ or, $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \big([(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a), \text{OFF}\big) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma](a)$ if $a \in [(T \cap Q) - S]_{\text{OFF}}$.
Case $a \in (S \cap T \cap Q)$,

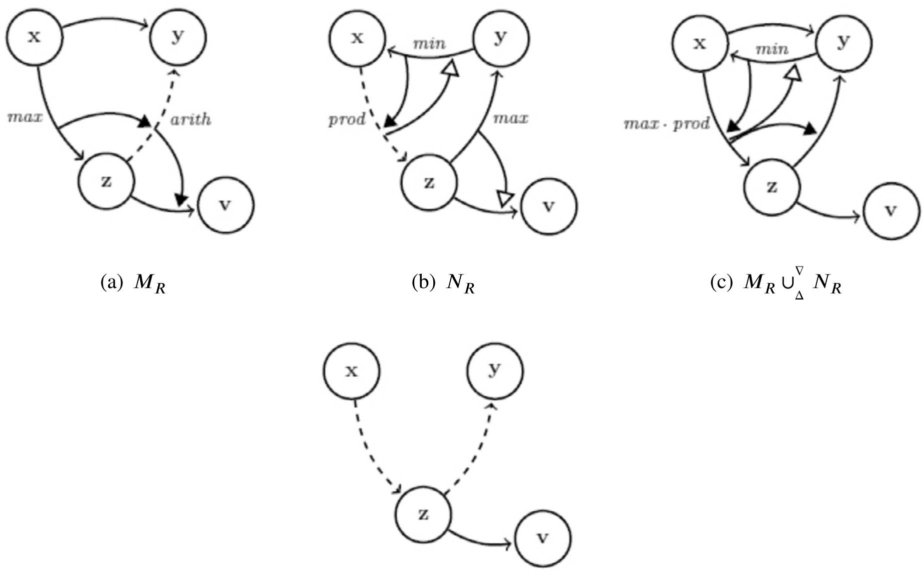
$$\begin{aligned}[\mu \cup_\Delta (\eta \cup_\Delta \gamma)]_1(a) &= \Delta\big(\mu_1(a), (\eta \cup_\Delta \gamma)_1(a)\big)\\ &= \Delta\big(\mu_1(a), \Delta\big(\eta_1(a), \gamma_1(a)\big)\big)\\ &= \Delta\big(\Delta\big(\mu_1(a), \eta_1(a)\big), \gamma_1(a)\big) \text{ (Associativity)}\\ &= \Delta\big((\mu \cup_\Delta \eta)_1(a), \gamma_1(a)\big)\\ &= [(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a)\end{aligned}$$

Then $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \big([(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a), \text{ON}\big) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a)$ if $a \in (S \cap T \cap Q)_{\text{ON}} \cup (S \cap T \cap Q)_{\neq}$ or, $[\mu \cup_\Delta (\eta \cup_\Delta \gamma)](a) = \big([(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a), \text{OFF}\big) = [(\mu \cup_\Delta \eta) \cup_\Delta \gamma]_1(a)$ if $a \in (S \cap T \cap Q)_{\text{OFF}}$.
Similarly, we prove $M \cap_\Delta (N \cap_\Delta P) = (M \cap_\Delta N) \cap_\Delta P$. □

For $RFRG$s, the intersection is not straightforward. It is possible that two operands contain the same arrow, but with different associated aggregations. In this case, we need to decide who will be the aggregation associated to the resulting arrow of the intersection. Before, let be the $RFRGs$ M_R and N_R. If we consider the set $(S_\rightarrow \cap T_\rightarrow)$, we will denote the sets

$$\big(S \cap T\big)^U_\rightarrow = \{a_i^0 \in (S_\rightarrow \cap T_\rightarrow);\ [\![a_i^0, a, \sigma]\!] \in M \cup_\Delta N\}$$

and



(a) M_R (b) N_R (c) $M_R \cup_{\Delta}^{\nabla} N_R$

(d) $M_R \cap_{\Delta}^{\nabla} N_R$

Fig. 9. Δ^{∇}-union and Δ^{∇}-intersection of M_R and N_R.

$$\big(S \cap T\big)^{I}_{\rightarrow} = \{a_i^0 \in (S_{\rightarrow} \cap T_{\rightarrow});\ [\![a_i^0, a, \sigma]\!] \in M \cap_{\Delta} N\},$$

for $\sigma \in \{\circ, \bullet\}$.

Definition 25. Let be the $RFRGs$ M_R and N_R. The structure $M_R \cup_{\Delta}^{\nabla} N_R = \langle M \cup_{\Delta} N, Ag^{\nabla}_{(M\cup_{\Delta} N)}\rangle$ s.t. the function $Ag^{\nabla}_{(M\cup_{\Delta} N)} : (S_{\rightarrow} \cup T_{\rightarrow}) \rightarrow A_M \cup A_N \cup \{\Pi_1 : [0,1]^3 \rightarrow [0,1]\}$, defined by

$$Ag^{\nabla}_{(M\cup_{\Delta} N)}(a_i^0) = \begin{cases} Ag_M(a_i^0), \text{ if } a_i^0 \in (S_{\rightarrow} - T_{\rightarrow}) \\ Ag_N(a_i^0), \text{ if } a_i^0 \in (T_{\rightarrow} - S_{\rightarrow}) \\ \Pi_1\Big(\nabla\big(Ag_M(a_i^0), Ag_N(a_i^0)\big), Ag_M(a_i^0), Ag_N(a_i^0)\Big), \text{ if } a_i^0 \in \big(S \cap T\big)^{U}_{\rightarrow} \end{cases}$$

is called Δ^{∇}**-union** of M_R and N_R.

The structure $M_R \cap_{\Delta}^{\nabla} N_R = \langle M \cap_{\Delta} N, Ag^{\nabla}_{(M\cap_{\Delta} N)}\rangle$ s.t. $Ag^{\nabla}_{(M\cap_{\Delta} N)} : (S_{\rightarrow} \cap T_{\rightarrow}) \rightarrow \{\Pi_1 : [0,1]^3 \rightarrow [0,1]\}$ is defined by

$$Ag^{\nabla}_{(M\cap_{\Delta} N)}(a_i^0) = \Pi_1\Big(\nabla\big(Ag_M(a_i^0), Ag_N(a_i^0)\big), Ag_M(a_i^0), Ag_N(a_i^0)\Big),$$

for $a_i^0 \in \big(S \cap T\big)^{I}_{\rightarrow}$, is called Δ^{∇}**-intersection of M_R and N_R.**

Example 26. Let be $\nabla(A_1, A_2)(\bar{x}) = A_1(\bar{x}) \cdot A_2(\bar{x})$ for $\bar{x} \in [0,1]^3$. The Fig. 9 shows two $RFRG$s M_R and N_R, together with $M_R \cup_{\Delta}^{\nabla} N_R$ and $M_R \cap_{\Delta}^{\nabla} N_R$.

The next proposition guarantees that the reconfiguration of an Δ^{∇}-union of $RFRG$s – for arrows contained in only one of the graphs – is equal to Δ^{∇}-union of its reconfigured components. Fig. 10 shows the case that the edge $[xy]$ has been crossed. For this example, consider $Ag_{M_E}([xy]) = product$ and $\Delta = max$.

The demonstration of the following Proposition will be done by using the previous patterns for connecting and disconnecting arrows. For $\sigma = \circ$ and $\sigma = \bullet$, we use OFF and ON, respectively. Succinctly, we will use OFF\ON for $\circ\backslash\bullet$.

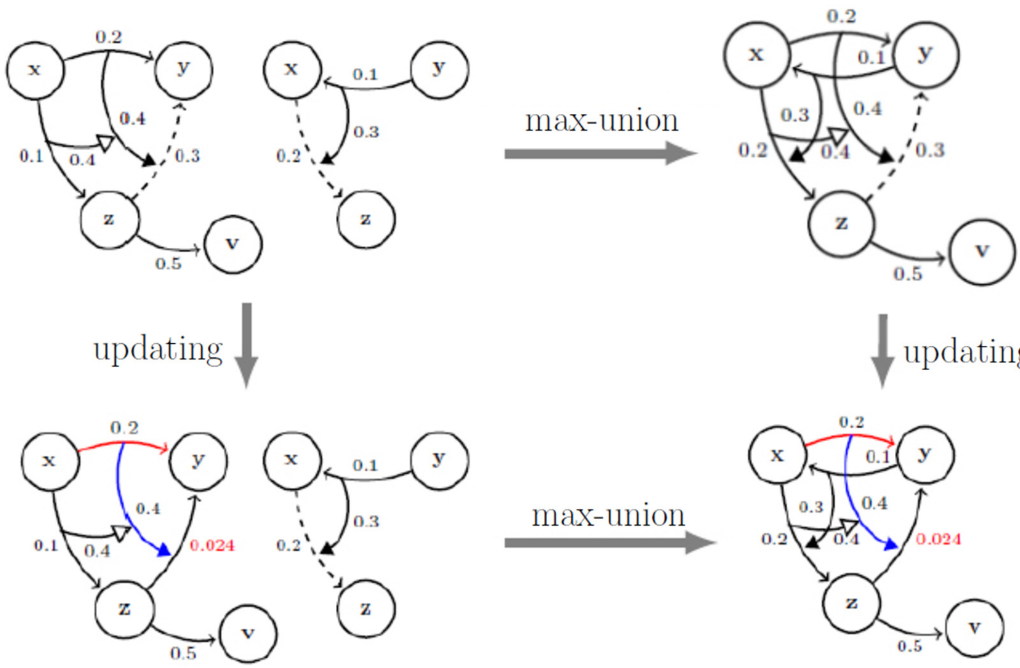


Fig. 10. Proposition 27.

Proposition 27. *Given $M_R \cup_{\Delta}^{\nabla} N_R$, an active arrow $a_i^0 \in S^0 - T^0$ $(T^0 - S^0)$ and $a \in S - T$ $(T - S)$, then:*

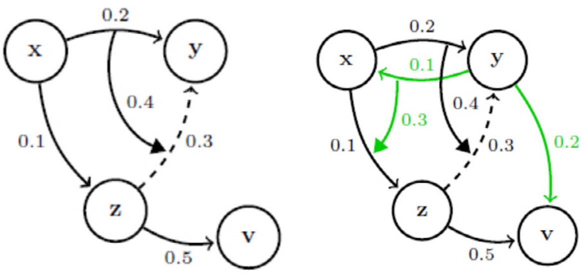
$$(\mu \cup_\Delta \eta)_{a_i^0}^{Ag_{(M \cup_\Delta N)}}(a) = \begin{cases} \mu(a), \text{ if } a_i^0 \in (S^0 - T^0)^*, a \in S - T \text{ and } [\![a_i^0, a, \sigma]\!] \notin (S \cup T)^*, \\ \eta(a), \text{ if } a_i^0 \in (T^0 - S^0)^*, a \in T - S \text{ and } [\![a_i^0, a, \sigma]\!] \notin (S \cup T)^*, \\ \left(Ag_M(a_i^0)\left(\mu_1(a_i^0), \mu_1([\![a_i^0, a, \sigma]\!]), \mu_1(a)\right), \text{OFF/ON}\right), \text{ if } a_i^0 \in (S^0 - T^0)^*, \\ a \in S - T \text{ and } [\![a_i^0, a, \sigma]\!] \in (S \cup T)^*, \\ \left(Ag_N(a_i^0)\left(\eta_1(a_i^0), \eta_1([\![a_i^0, a, \sigma]\!]), \eta_1(a)\right), \text{OFF/ON}\right), \text{ if } a_i^0 \in (T^0 - S^0)^*, \\ a_i^0 \in T - S \text{ and } [\![a_i^0, a, \sigma]\!] \in (S \cup T)^*, \end{cases}$$

for $\sigma = \circ/\bullet$.

Proof. Indeed,

- Case $[\![a_i^0, a, \sigma]\!] \notin (S \cup T)^*$, $(\mu \cup_\Delta \eta)_{a_i^0}^{Ag_{M \cup_\Delta N}}(a) \overset{\text{def}}{=} (\mu \cup_\Delta \eta)(a)$ and the result follows.
- Case $[\![a_i^0, a, \sigma]\!] \in (S \cup T)^*$:
$(\mu \cup_\Delta \eta)_{a_i^0}^{Ag_{(M \cup_\Delta N)}}(a) = \left(Ag^{\nabla}_{(M \cup_\Delta N)}(a_i^0)\left((\mu \cup_\Delta \eta)_1(a_i^0), (\mu \cup_\Delta \eta)_1([\![a_i^0, a, \sigma]\!]), (\mu \cup_\Delta \eta)_1(a)\right), \text{OFF/ON}\right) \overset{\text{def}}{=}$

$$\overset{\text{def}}{=} \begin{cases} \left(Ag^{\nabla}_{(M \cup_\Delta N)}(a_i^0)\left(\mu_1(a_i^0), \mu_1([\![a_i^0, a, \sigma]\!]), \mu_1(a)\right), \text{OFF/ON}\right), \text{ if } a \in S - T \text{ and } a_i^0 \in (S^0 - T^0)^* \\ \left(Ag^{\nabla}_{(M \cup_\Delta N)}(a_i^0)\left(\eta_1(a_i^0), \eta_1([\![a_i^0, a, \sigma]\!]), \eta_1(a)\right), \text{OFF/ON}\right), \text{ if } a \in T - S \text{ and } a_i^0 \in (T^0 - S^0)^* \end{cases}$$



(a) M (b) M_E

Fig. 11. $RSFG$s M and M_E.

$$\stackrel{\text{def}}{=}\begin{cases}\Big(Ag_M(a_i^0)\Big(\mu_1(a_i^0),\mu_1([\![a_i^0,a,\sigma]\!]),\mu_1(a)\Big),\text{OFF/ON}\Big), \textit{if } a\in S-T \text{ and } a_i^0\in(S^0-T^0)^*\\ \Big(Ag_N(a_i^0)\Big(\eta_1(a_i^0),\eta_1([\![a_i^0,a,\sigma]\!]),\eta_1(a)\Big),\text{OFF/ON}\Big), \textit{if } a\in T-S \text{ and } a_i^0\in(T^0-S^0)^*,\end{cases}$$

for $\sigma=\circ/\bullet$. □

The reconfiguration of the Δ-union of two $RFSG$s, for arrows in intersection, can be not equivalent to the Δ-union of their reconfigurations due the action of Δ. For these arrows, after the Δ-union, fuzzy values different from the originals will be assigned and this fact promotes the non-validity of the above proposition.

The Proposition 27 it will also not be valid for arrows from the Δ-intersection between two graphs. Consider the min-intersection of the Fig. 8(b) which cannot even be updated.

4.2. Extension of $RFSGs$ and $RFRGs$

This section introduces an operation which enables to reuse a model in order to build a more complex one. An extension of a system $M=\langle W,\mu:S\to[0,1]\times\{\circ,\bullet\}\rangle$ involves a non-empty finite set of new arrows (S_E), a membership function ($\varepsilon_E:S_E\to[0,1]\times\{\text{ON},\text{OFF}\}$), and a finite set (possibly empty) of new nodes (W_E) to be connected to the original graph. For example, considering the $RFSG$ M in Fig. 11(a), the Fig. 11(b) shows its extension with $W_E=\emptyset$, $S_E=\{[yx],[yv],[\![[yx],[xz],\bullet]\!]\}$, a membership function ε_E s.t. $\varepsilon_E([yx])=(0.1,\text{ON})$, $\varepsilon_E([yv])=(0.2,\text{ON})$ and $\varepsilon_E([\![[yx],[xz],\bullet]\!])=(0.3,\text{ON})$.

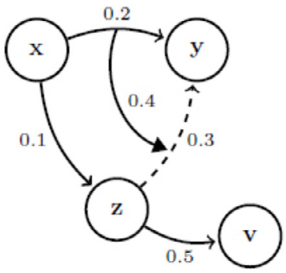
Another extension for M can be viewed in Fig. 12(b). A set with a new set of nodes $W_E=\{t,s\}$ and a set of new arrows $S_E=\{[yt],[tv],[vs],[\![[yt],[tv],\circ]\!],[\![[xy],[yt],\circ]\!],[\![[vs],[zy],\bullet]\!]\}$ together with the membership function ε_E s.t. $\varepsilon_E([yt])=(0.2,\text{ON})$, $\varepsilon_E([tv])=(0.1,\text{ON})$, $\varepsilon_E([\![[xy],[yt],\circ]\!])=(0.6,\text{ON})$, $\varepsilon_E([vs])=(0.1,\text{ON})$, $\varepsilon_E([\![[vs],[zy],\bullet]\!])=(0.4,\text{ON})$ and $\varepsilon_E([\![[yt],[tv],\circ]\!])=(0.6,\text{ON})$. Taking into account the set of arrows $S=\{[xy],[xz],[zy],[zv],[\![[xy],[zy],\bullet]\!]\}$, the extension M_E has the set of nodes $W\cup W_E$, the set of arrows $S\cup S_E$. For all $w_e\in W_E$, there is a $w\in W\cup W_E$ with $[ww_e]\in S_E$ or $[w_ew]\in S_E$ and there is a $w'\in W_E$ such $[w''w']\in S_E$ or $[w'w'']\in S_E$, with $w''\in W$. The next definition formalizes this situation.

Definition 28. Let be a $RFSG$ $M=\langle W,\mu:S\to[0,1]\times\{\circ,\bullet\}\rangle$ and a structure $E=\langle W_E,S_E,\varepsilon_E\rangle$ s.t.,

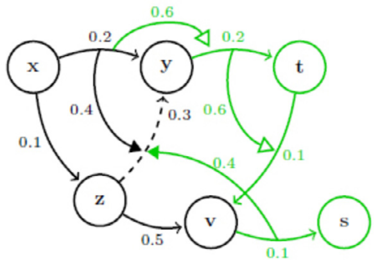
- W_E is a finite (possible empty) **set of new states** which $W\cap W_E=\emptyset$;
- $S_E=\bigcup_{n\in\mathbb{N}}S_E^n$ is a **set of new arrows** s.t.

$$\begin{cases}{S_E}^0\subseteq\Big(\bigcup_{I,J\in\{W_E,W\}}(I\times J)\Big),\\ {S_E}^{n+1}\subseteq\bigcup_{n\in\mathbb{N}}\Big((S^0\cup S_E^0)\times(S^n\cup S_E^n)\times\{\circ,\bullet\}\Big),\end{cases}$$

with $S_E\cap S=\emptyset$ and, for $n\geq 1$, $\sigma_0,\sigma_1,\sigma_2,\sigma_3\in\{\circ,\bullet\}$:



(a) M



(b) M_E

Fig. 12. $RSFG$s M and M_E.

Table 1
Fuzzy values of relation S^0 - zero-order arrows in W.

	x	y	z	v
x		**0.2**	**0.1**	
y				
z		**0.3**		**0.5**
v				

Table 2
Fuzzy values of relation $S^0 \cup S^0_E$ - zero-order arrows in $W \cup W_E$ in Fig. 11(b).

	x	y	z	v
x		**0.2**	**0.1**	
y	**0.1**			**0.2**
z		**0.3**		**0.5**
v				

  - $[\![a_i^0, a, \sigma_0]\!] \notin S^n$ or $[\![a_i^0, a, \sigma_1]\!] \notin S^n_E$,
  - If $[\![a_i^0, a, \sigma_2]\!] \in S^n_E$ with $a_i^0 \in S$, then $\exists b \in S \cup T, b \neq a$ and $[\![a_i^0, b, \sigma_3]\!] \in S^n$.
- $\varepsilon_E : S_E \to [0, 1] \times \{\text{ON}, \text{OFF}\}$ is a **membership function.**

The **extension of** M **based on the structure** $\langle W_E, S_E, \varepsilon_E\rangle$ **is the** $RFSG$ $M_E = \langle W \cup W_E;\ \mu_E : S \cup S_E \to [0, 1] \times \{\text{ON}, \text{OFF}\}\rangle$ **with**

$$\mu_E(a) = \begin{cases} \varepsilon_E(a), \textbf{case } a \in S_E, \\ \mu(a), \textbf{case } a \in S. \end{cases} \tag{8}$$

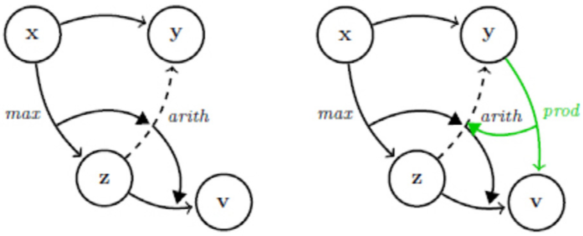
Example 29. Given the set of states in Fig. 11(a), Table 1 expresses the fuzzy values of the relation S^0.
Table 2 show the fuzzy values of $S^0 \cup S^0_E$. Here, we have $W_E = \emptyset$ and $S^0_E = \{[yx], [yv]\}$.
Table 3 shows the fuzzy values of zero-order arrows in Fig. 12(b). Here, we have $W_E = \{t, s\}$ and $S_E = \{[yt], [tv], [vs], [\![[yt], [tv], \circ]\!], [\![[xy], [yt], \circ]\!], [\![[vs], [zy], \bullet]\!]\}$.
The tables for the high-order relations follow similarly.

In order to define the extension of a $RFRG$ M_R, we need to introduce the following notation:

- A finite and non-empty set of aggregations $A_{S_E} = \{A_1, A_2, ..., A_s : [0, 1]^3 \to [0, 1]\}$.

Table 3
Fuzzy values of relation $S^0 \cup S_E^0$ - zero-order arrows in $W \cup W_E$ in Fig. 12(b).

	x	y	z	v	t	s
x		**0.2**	**0.1**			
y					**0.2**	
z		**0.3**		**0.5**		
v						**0.1**
t				**0.1**		
s						



(a) M_R (b) $(M_R)_E$

Fig. 13. M_R and associated $(M_R)_E$.

- An function $Ag_{S_E} : S_{E_\to} \to A_{S_E}$.

Definition 30. Let $M = \langle W, \mu : S \to [0, 1] \times \{\text{ON}, \text{OFF}\}\rangle$ be a $RFSG$ and $M_E = \langle W \cup W_E;\ \mu_E : S_E \cup S \to [0, 1] \times \{\text{ON}, \text{OFF}\}\rangle$ an extension of M. The extension of the $RFRG$ $M_R = \langle M, Ag_M\rangle$ is the pair $(M_R)_E = \langle M_E, Ag_{M_E} : S_{E_\to} \cup S_\to \to A_{S_E} \cup A_M\rangle$ s.t.

$$Ag_{M_E}(a_i^0) = \begin{cases} Ag_M(a_i^0), & \text{if } a_i^0 \in S_\to, \\ Ag_{S_E}(a_i^0), & \text{if } a_i^0 \in S_{E_\to}. \end{cases}$$

Example 31. Consider the $RFRG$ M_R in Fig. 9(a). If $S_E = \{[yv], [\![[yv], [zy], \bullet]\!]\}$, the set $A_{S_E} = \{prod\}$ and the application $Ag_{S_E} : S_E \to A_{S_E}$ s.t. $Ag_{S_E}([yv]) = prod$, we have the $RFRG$ $(M_R)_E$ in Fig. 13(b).

$RFSG$s have at most one ternary aggregation function A associated. Since the extension of a $RFSG$ is also a $RFSG$, then it will have at most one aggregation associated to its arrows. So we define the reconfiguration of extended $RFSG$s in the following way.

Proposition 32. *The* ***reconfiguration of*** *M_E, based on the aggregation A after crossing the active arrow $a_i^0 \in S \cup S_E$ is the $RFSG$*

$$(M_E)^{A_s}_{a_i^0} = \langle W \cup W_E; (\mu_E)^A_{a_i^0} : S \cup S_E \to [0, 1] \times \{\text{ON}, \text{OFF}\}\rangle$$

s.t. $(\mu_E)^A_{a_i^0}(a) =$

$$\begin{cases} \mu(a), \text{ if } a \in S \text{ and } [\![a_i^0, a, \sigma]\!] \notin (S \cup S_E)^* \\ \varepsilon_E(a), \text{ if } a \in S_E \text{ and } [\![a_i^0, a, \sigma]\!] \notin (S \cup S_E)^* \\ \Big(A\big(\mu_1(a_i^0), \mu_1([\![a_i^0, a, \sigma]\!]), \mu_1(a)\big), \text{OFF/ON}\Big), \text{ if } a \in S \text{ and } a_i^0, [\![a_i^0, a, \sigma]\!] \in S^* \\ \Big(A\big(\varepsilon_{E1}(a_i^0), \varepsilon_{E1}([\![a_i^0, a, \sigma]\!]), \varepsilon_{E1}(a)\big), \text{OFF/ON}\Big), \text{ if } a \in S_E \text{ and } a_i^0, [\![a_i^0, a, \sigma]\!] \in S_E^* \\ \Big(A\big(\mu_1(a_i^0), \varepsilon_{E1}([\![a_i^0, a, \sigma]\!]), \mu_1(a)\big), \text{OFF/ON}\Big), \text{ if } a \in S, a_i^0 \in S^* \text{ and } [\![a_i^0, a, \sigma]\!] \in S_E^* \\ \Big(A\big(\varepsilon_{E1}(a_i^0), \varepsilon_{E1}([\![a_i^0, a, \sigma]\!]), \mu_1(a)\big), \text{OFF/ON}\Big), \text{ if } a \in S \text{ and } a_i^0, [\![a_i^0, a, \sigma]\!] \in S_E^* \\ \Big(A\big(\mu_1(a_i^0), \varepsilon_{E1}([\![a_i^0, a, \sigma]\!]), \varepsilon_{E1}(a)\big), \text{OFF/ON}\Big), \text{ if } a_i^0 \in S^*, a \in S_E \text{ and } [\![a_i^0, a, \sigma]\!] \in S_E^* \end{cases}$$

for $a \in S \cup S_E$ and $\sigma = \circ/\bullet$.

Proof. Indeed,

- Case $a \in S$ and $[\![a_i^0, a, \sigma]\!] \notin (S \cup S_E)^*$: $(\mu_E)^A_{a_i^0}(a) \stackrel{\text{def}}{=} \mu_E(a) = \mu(a)$.
- Case $a \in S_E$ and $[\![a_i^0, a, \sigma]\!] \notin (S \cup S_E)^*$: $(\mu_E)^A_{a_i^0}(a) \stackrel{\text{def}}{=} \mu_E(a) = \varepsilon_E(a)$.
- Case $a \in S$ and $a_i^0, [\![a_i^0, a, \sigma]\!] \in S^*$: $(\mu_E)^A_{a_i^0}(a) \stackrel{\text{def}}{=} \Big(A\big(\mu_{E1}(a_i^0), \mu_{E1}([\![a_i^0, a, \sigma]\!]), \mu_{E1}(a)\big), \text{OFF/ON}\Big) = \Big(A\big(\mu_1(a_i^0), \mu_1([\![a_i^0, a, \sigma]\!]), \mu_1(a)\big), \text{OFF/ON}\Big)$, for $\sigma = \circ/\bullet$.
- Case $a \in S_E$ and $a_i^0, [\![a_i^0, a, \sigma]\!] \in S_E^*$: $(\mu_E)^A_{a_i^0}(a) \stackrel{\text{def}}{=} \Big(A\big(\mu_{E1}(a_i^0), \mu_{E1}([\![a_i^0, a, \sigma]\!]), \mu_{E1}(a)\big), \text{OFF/ON}\Big) = \Big(A\big(\varepsilon_{E1}(a_i^0), \varepsilon_{E1}([\![a_i^0, a, \sigma]\!]), \varepsilon_{E1}(a)\big), \text{OFF/ON}\Big)$, for $\sigma = \circ/\bullet$.
- Case $a \in S, a_i^0 \in S^*$ and $[\![a_i^0, a, \sigma]\!] \in S_E^*$: $(\mu_E)^A_{a_i^0}(a) \stackrel{\text{def}}{=} \Big(A\big(\mu_{E1}(a_i^0), \mu_{E1}([\![a_i^0, a, \sigma]\!]), \mu_{E1}(a)\big), \text{OFF/ON}\Big) = \Big(A\big(\mu_1(a_i^0), \varepsilon_{E1}([\![a_i^0, a, \sigma]\!]), \mu_1(a)\big), \text{OFF/ON}\Big)$, for $\sigma = \circ/\bullet$.
- Case $a \in S$ and $a_i^0, [\![a_i^0, a, \sigma]\!] \in S_E^*$: $(\mu_E)^A_{a_i^0}(a) \stackrel{\text{def}}{=} \Big(A\big(\mu_{E1}(a_i^0), \mu_{E1}([\![a_i^0, a, \sigma]\!]), \mu_{E1}(a)\big), \text{OFF/ON}\Big) = \Big(A\big(\varepsilon_{E1}(a_i^0), \varepsilon_{E1}([\![a_i^0, a, \sigma]\!]), \mu_1(a)\big), \text{OFF/ON}\Big)$, for $\sigma = \circ/\bullet$.
- Case $a_i^0 \in S^*, a \in S_E$ and $[\![a_i^0, a, \sigma]\!] \in S_E^*$: $(\mu_E)^A_{a_i^0}(a) \stackrel{\text{def}}{=} \Big(A\big(\mu_{E1}(a_i^0), \mu_{E1}([\![a_i^0, a, \sigma]\!]), \mu_{E1}(a)\big), \text{OFF/ON}\Big) = \Big(A\big(\mu_1(a_i^0), \varepsilon_{E1}([\![a_i^0, a, \sigma]\!]), \varepsilon_{E1}(a)\big), \text{OFF/ON}\Big)$, for $\sigma = \circ/\bullet$. $\square$

Remark 6. Observe that the cases in which $[\![a_i^0, a, \sigma]\!] \in S$ and $a \in S_E$ do not make sense for $a \in S \cup S_E = \emptyset$.

Proposition 33. *Consider $(M_R)_E$, an active zero-order arrow $a_i^0 \in S^0 \cup S_E^0$ and $a \in (S \cup S_E)$. Let the RFRG $\big[(M_R)_E\big]_{a_i^0}^{Ag_{M_E}} = \big\langle (M_E)_{a_i^0}^{Ag_{M_E}}, Ag_{M_E} \big\rangle$, where $(M_E)_{a_i^0}^{Ag_{M_E}} = \big\langle W \cup W_E, (\mu_E)_{a_i^0}^{Ag_{M_E}} : S \cup S_E \to [0,1] \times \{\text{ON}, \text{OFF}\} \big\rangle$. The application $(\mu_E)_{a_i^0}^{Ag_{M_E}}$ is defined by:*

$$(\mu_E)_{a_i^0}^{Ag_{M_E}}(a)=\begin{cases} \mu(a),\ \textit{if}\ a\in S\ \textit{and}\ [\![a_i^0,a,\sigma]\!]\notin(S\cup S_E)^* \\ \varepsilon(a),\ \textit{if}\ a\in S_E\ \textit{and}\ [\![a_i^0,a,\sigma]\!]\notin(S\cup S_E)^* \\ \left(Ag_M(a_i^0)\big(\mu_1(a_i^0),\mu_1([\![a_i^0,a,\sigma]\!]),\mu_1(a)\big),\text{OFF/ON}\right),\ \textit{if}\ a\in S\ \text{and}\ a_i^0,[\![a_i^0,a,\sigma]\!]\in S^* \\ \left(Ag_{S_E}(a_i^0)\big(\varepsilon_{E1}(a_i^0),\varepsilon_{E1}([\![a_i^0,a,\sigma]\!]),\varepsilon_{E1}(a)\big),\text{OFF/ON}\right),\ \textit{if}\ a\in S_E\ \text{and}\ a_i^0,[\![a_i^0,a,\sigma]\!]\in S_E^* \\ \left(Ag_{S_E}(a_i^0)\big(\varepsilon_{E1}(a_i^0),\varepsilon_{E1}([\![a_i^0,a,\sigma]\!]),\mu_1(a)\big),\text{OFF/ON}\right),\ \textit{if}\ a\in S\ \text{and}\ a_i^0,[\![a_i^0,a,\sigma]\!]\in S_E^* \\ \left(Ag_M(a_i^0)\big(\mu_1(a_i^0),\varepsilon_{E1}([\![a_i^0,a,\sigma]\!]),\mu_1(a)\big),\text{OFF/ON}\right),\ \textit{if}\ a\in S, a_i^0\in S^*\ \text{and}\ [\![a_i^0,a,\sigma]\!]\in S_E^* \\ \left(Ag_M(a_i^0)\big(\mu_1(a_i^0),\varepsilon_{E1}([\![a_i^0,a,\sigma]\!]),\varepsilon_{E1}(a)\big),\text{OFF/ON}\right),\ \textit{if}\ a\in S_E, a_i^0\in S^*\ \text{and}\ [\![a_i^0,a,\sigma]\!]\in S_E^*, \end{cases}$$

for $\sigma=\circ\backslash\bullet$.

The $RFRG\ \big[(M_R)_E\big]_{a_i^0}^{Ag_{M_E}}$ *is the* ***reconfiguration of*** $(M_R)_E$ ***based in*** Ag_{M_E} ***after*** a_i^0 ***has been crossed.***

Proof. Indeed,

- Case $a\in S$ and $[\![a_i^0,a,\sigma]\!]\notin(S\cup S_E)^*$: $(\mu_E)_{a_i^0}^{Ag_{M_E}}(a)\overset{\text{def}}{=}\mu_E(a)=\mu(a)$.
- Case $a\in S_E$ and $[\![a_i^0,a,\sigma]\!]\notin(S\cup S_E)^*$: $(\mu_E)_{a_i^0}^{Ag_{M_E}}(a)\overset{\text{def}}{=}\mu_E(a)=\varepsilon(a)$.
- Case $a\in S$ and $a_i^0,[\![a_i^0,a,\sigma]\!]\in S^*$:
$(\mu_E)_{a_i^0}^{Ag_{M_E}}(a)\overset{\text{def}}{=}\left(Ag_{M_E}(a_i^0)\big(\mu_{E1}(a_i^0),\mu_{E1}([\![a_i^0,a,\sigma]\!]),\mu_{E1}(a)\big),\text{OFF/ON}\right)=$
$\left(Ag_M(a_i^0)\big(\mu_1(a_i^0),\mu_1([\![a_i^0,a,\sigma]\!]),\mu_1(a)\big),\text{OFF/ON}\right)$, for $\sigma=\circ/\bullet$.
- Case $a\in S_E$ and $a_i^0,[\![a_i^0,a,\sigma]\!]\in S_E^*$:
$(\mu_E)_{a_i^0}^{Ag_{M_E}}(a)\overset{\text{def}}{=}\left(Ag_{M_E}(a_i^0)\big(\mu_{E1}(a_i^0),\mu_{E1}([\![a_i^0,a,\sigma]\!]),\mu_{E1}(a)\big),\text{OFF/ON}\right)=$
$\left(Ag_{S_E}(a_i^0)\big(\varepsilon_{E1}(a_i^0),\varepsilon_{E1}([\![a_i^0,a,\sigma]\!]),\varepsilon_{E1}(a)\big),\text{OFF/ON}\right)$, for $\sigma=\circ/\bullet$.
- Case $a\in S$ and $a_i^0,[\![a_i^0,a,\sigma]\!]\in S_E^*$:
$(\mu_E)_{a_i^0}^{Ag_{M_E}}(a)\overset{\text{def}}{=}\left(Ag_{M_E}(a_i^0)\big(\mu_{E1}(a_i^0),\mu_{E1}([\![a_i^0,a,\sigma]\!]),\mu_{E1}(a)\big),\text{OFF/ON}\right)=$
$\left(Ag_{S_E}(a_i^0)\big(\varepsilon_{E1}(a_i^0),\varepsilon_{E1}([\![a_i^0,a,\sigma]\!]),\mu_1(a)\big),\text{OFF/ON}\right)$, for $\sigma=\circ/\bullet$.
- Case $a\in S$, $a_i^0\in S^*$ and $[\![a_i^0,a,\sigma]\!]\in S_E^*$:
$(\mu_E)_{a_i^0}^{Ag_{M_E}}(a)\overset{\text{def}}{=}\left(Ag_{M_E}(a_i^0)\big(\mu_{E1}(a_i^0),\mu_{E1}([\![a_i^0,a,\sigma]\!]),\mu_{E1}(a)\big),\text{OFF/ON}\right)=$
$\left(Ag_M(a_i^0)\big(\mu_1(a_i^0),\varepsilon_{E1}([\![a_i^0,a,\sigma]\!]),\varepsilon_{E1}(a)\big),\text{OFF/ON}\right)$, for $\sigma=\circ/\bullet$.
- Case $a\in S_E$, $a_i^0\in S^*$ and $[\![a_i^0,a,\sigma]\!]\in S_E^*$:
$(\mu_E)_{a_i^0}^{Ag_{M_E}}(a)\overset{\text{def}}{=}\left(Ag_{M_E}(a_i^0)\big(\mu_{E1}(a_i^0),\mu_{E1}([\![a_i^0,a,\sigma]\!]),\mu_{E1}(a)\big),\text{OFF/ON}\right)=$
$\left(Ag_M(a_i^0)\big(\mu_1(a_i^0),\varepsilon_{E1}([\![a_i^0,a,\sigma]\!]),\varepsilon_{E1}(a)\big),\text{OFF/ON}\right)$, for $\sigma=\circ/\bullet$. □

The following proposition guarantees that, given a $RFRG$ $(M_R)_E$, if an active zero-order arrow $a_i^0 \in S$ has been crossed and it is associated with an arrow $a \in S$ by a high-order arrow $[\![a_i^0, a, \sigma]\!] \in S$, then the reconfiguration of $(M_R)_E$ coincides with the extension of the reconfiguration of M_R after crossing a_i^0.

Proposition 34. *Let be* $(M_R)_E = \langle M_E, Ag_{M_E}\rangle$, $a_i^0 \in S^*$, $a \in S$, *and* $[\![a_i^0, a, \sigma]\!] \in S^*$ *then,*

$$\big[(M_R)_E\big]_{a_i^0}^{Ag_{M_E}} = \big[(M_R)_{a_i^0}^{Ag_M}\big]_E$$

Proof. According to Proposition 33, $\big[(M_R)_E\big]_{a_i^0}^{Ag_{M_E}} = \big\langle (M_E)_{a_i^0}^{Ag_{M_E}}, Ag_{M_E}\big\rangle$ where

$$(M_E)_{a_i^0}^{Ag_{M_E}} = \big\langle W \cup W_E; (\mu_E)_{a_i^0}^{Ag_{M_E}} : S \cup S_E \to [0,1] \times \{\text{ON}, \text{OFF}\}\big\rangle.$$

Similarly, according to Definition 28, $\big[(M_R)_{a_i^0}^{Ag_M}\big]_E = \big\langle (M_{a_i^0}^{Ag_M})_E, Ag_{M_E}\big\rangle$ s.t.

$$(M_{a_i^0}^{Ag_M})_E = \big\langle W \cup W_E; \big[(\mu)_{a_i^0}^{Ag_M}\big]_E : S \cup S_E \to [0,1] \times \{\text{ON}, \text{OFF}\}\big\rangle.$$

So, for $a \in S$, $a_i^0 \in S^*$ and $[\![a_i^0, a, \sigma]\!] \in S^*$, $(\mu_E)_{a_i^0}^{Ag_{M_E}} = \big[(\mu)_{a_i^0}^{Ag_M}\big]_E$. Indeed, according to Definition 28,

$$\begin{aligned}
\big[(\mu)_{a_i^0}^{Ag_M}\big]_E(a) &= (\mu)_{a_i^0}^{Ag_M}(a) \\
&= \begin{cases} \mu(a), \text{ if } [\![a_i^0, a, \sigma]\!] \notin S^*; \\ \Big(Ag_M(a_i^0)\big(\mu_1(a), \mu_1(a_i^0), \mu_1([\![a_i^0, a, \bullet]\!])\big), \text{ON}\Big), \text{ if } [\![a_i^0, a, \bullet]\!] \in S^*. \\ \Big(Ag_M(a_i^0)\big(\mu_1(a), \mu_1(a_i^0), \mu_1([\![a_i^0, a, \circ]\!])\big), \text{OFF}\Big), \text{ if } [\![a_i^0, a, \circ]\!] \in S^*. \end{cases} \\
&= (\mu_E)_{a_i^0}^{Ag_{M_E}}(a)
\end{aligned}$$

and the result follows. □

Fig. 14 shows a diagram with the interpretation of the Proposition 34 in case the edge $[xz]$ is crossed. For this example, consider $Ag_{M_E}([xz]) = product$.

5. Coupling of auxiliary mechanisms in a tank level control system

Some industrial processes use tanks to control the level of fluids. The modeling of a system of such tanks and logic controllers is an important issue, since it provides an understanding of the current scenario of the system which induces some benefits, such as: increase of productivity and the prevention of accidents [8]. The extension of a tank system can be carried out by coupling auxiliary devices, such as valves, temperature meters, etc. In this section, we show how use the extension of $RFRG$s can be used to model the coupling of auxiliary mechanisms to a system of a tank level control.

Fig. 15(a) shows RFSGs which models a system such that a scheme where a tank control system is built with three signal transmitters $\{ST_1, ST_2, ST_3\}$, two pumps $\{P_1, P_2\}$ and a channel for fluid inlet called $START$. The dynamics of the system works as follows:

- **Fluid level increasing**: At $START$ the fluid starts to be inserted into the tank while pumps P_1 and P_2 are on standby receiving the minimum electric current. When the fluid level triggers ST_2, P_1 receives an increment of electric current and is activated. If the fluid level continues to rise and trigger ST_3, the pump P_2 receives an increment of electric current and is also activated.
- **Fluid level decreasing**: When the fluid level is maximum, the pumps P_1 and P_2 are activated. When the fluid level decreases, the ST_3 is triggered and P_2 goes to standby with the decreasing of its electric current. If the fluid level continues to decrease, ST_2 is triggered and P_1 also goes to standby with a decreasing of its electric current.

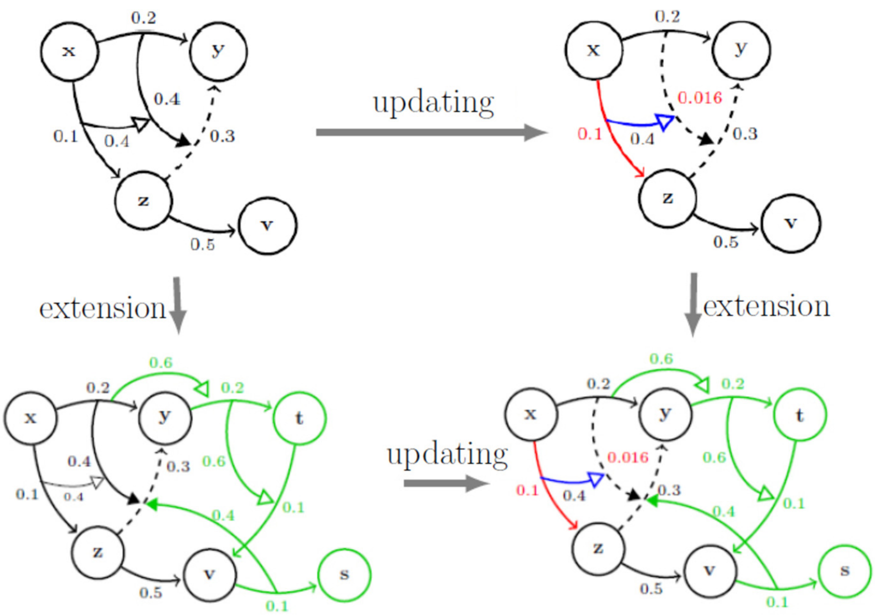
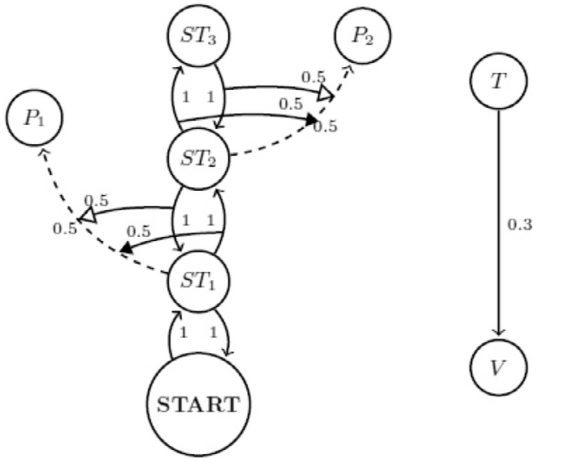


Fig. 14. Proposition 34.



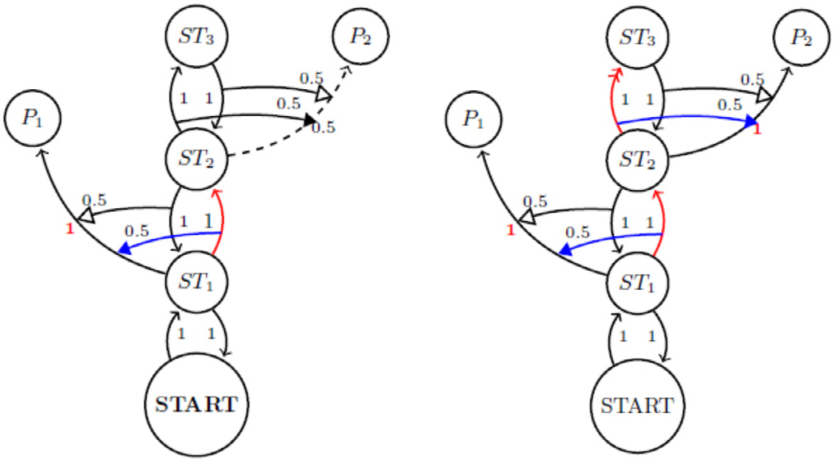
(a) M (b) N

Fig. 15. M is a tank control system and N is an auxiliary system.

The signal transmitter receives the pressure difference of two points with different heights and converts it into a proportional electrical signal. This electric signal is sent to pumps [8].

Fig. 15(b) shows a $RFSG$ which models a monitoring system composed of a valve V and a timer T. The dynamics of the system works as follows:

- **When coupled to a tank system:** The system N (Fig. 15(b)) is activated when the fluid reaches the maximum level. In this condition, the timer is activated and count t minutes for the fluid level to start to drop. If that doesn't happen, it sends an electrical signal to the valve that blocks the fluid inlet, until the decreasing of the level of the fluid begins to be detected.



(a) $M_R^{[ST_1\ ST_2]}$ (b) $M_R^{[ST_1\ ST_2][ST_2\ ST_3]}$

Fig. 16. Reconfigurations of M_R.

For each zero-order arrow of graphs M, we associate aggregations as follows:

$$Ag_M([ST_1\ ST_2]) = Ag_M([ST_2\ ST_3]) = max$$

and

$$Ag_M([ST_3\ ST_2]) = Ag_M([ST_2\ ST_1]) = min.$$

This choice was made in a convenient way so that whenever the fluid level increases, the aggregation of the fuzzy values is greater than the immediately previous value (representing the increase in the transmitted electric current). In the same way, whenever the fluid decreases, the result of the aggregation of the fuzzy values is less than the immediately previous values (representing the decrease of the transmitted electric current).

The systems in Fig. 15(a) and Fig. 15(b) are modeled by the $RFRGs$: $M_R = \langle M, Ag_M\rangle$ and $N_R = \langle N, Ag_N\rangle$ respectively. Fig. 16 shows the reconfiguration of M_R after the arrows $[ST_1\ ST_2]$ and the sequence of arrows $[ST_1\ ST_2][ST_2\ ST_3]$ have been crossed.

In order to increase the control over the system M_R, we extend it with the new nodes $W_E = \{T, V\}$ and the new arrows:

$$S_E = \Big\{[ST_3\ T], [T\ V], [ST_3\ ST_3], [\![[T\ V], [START\ ST_1], \circ]\!], [\![[ST_3\ ST_3], [ST_3\ T], \bullet]\!], \\ [\![[ST_3\ ST_3], [\![[ST_3\ ST_3], [ST_3\ T], \bullet]\!], \bullet]\!], [\![[ST_3\ ST_3], [\![[ST_3\ ST_3], [\![[ST_3\ ST_3], [ST_3\ T], \bullet]\!], \bullet]\!], \bullet]\!] \Big\}$$

s.t. $Ag_{S_E}([ST_3\ ST_3]) = Ag_{S_E}([ST_3\ T]) = max$ and $Ag_{S_E}([T\ V]) = min$. The resulting graph allows emergency actions to be taken on the system M_R, such as the closing of the fluid inlet by a valve in case of delay for the beginning of the degrowth of the fluid level.

Fig. 17 shows $(M_R)_E$. This graph represents the coupling of a control system in the pre-existing control system for tank level M_R.

Observe how the resulting extension works *after* the following arrows $[START\ ST_1]$, $[ST_1\ ST_2]$, $[ST_2\ ST_3]$ and $[ST_3\ ST_3]$ are crossed.

Step 1: $[START\ ST_1]$ has been crossed (Fig. 18(a)). In this case, the fluid starts to be inserted into the tank until it reaches ST_1. Pumps P_1 and P_2 are on standby.

Step 2: $[ST_1\ ST_2]$ has been crossed (Fig. 18(b)). The fluid level inside the tank continues to fill up to the ST_2. Pump P_1 is activated and the drainage process begins. In this case, $(\mu_E)^{max}_{[ST_1\ ST_2]}\big([ST_1\ P_1]\big) = \big(max(1, 0.5, 0.5), \text{ON}\big) = (1, \text{ON})$.

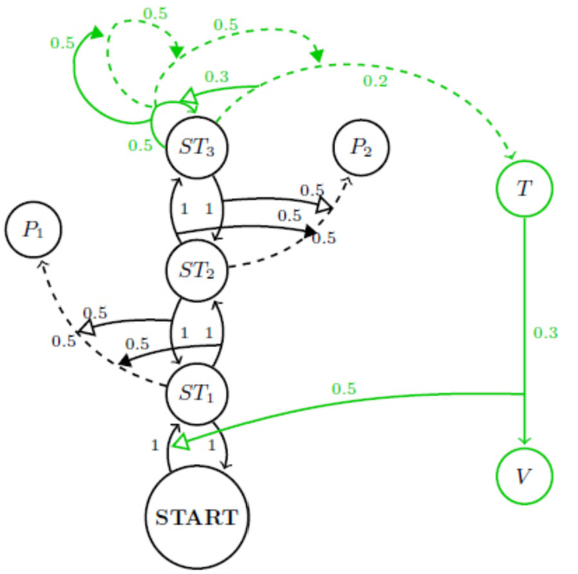


Fig. 17. $(M_R)_E$.

Step 3: $[ST_2\ ST_3]$ has been crossed (Fig. 18(c)). The fluid level inside the tank continues to fill up to the ST_3. Pump P_2 is activated and, together with the pump P_1, the drainage process continues. In this case, $(\mu_E)^{max}_{[ST_2\ ST_3]}([ST_2\ P_2]) = \big(max(1, 0.5, 0.5), \text{ON}\big) = (1, \text{ON})$.

Step 4: $[ST_3\ ST_3]$ has been crossed once a time (Fig. 18(d)). The timer action starts and a time interval is counted to start decreasing the fluid level. In this case, $(\mu_E)^{max}_{[ST_3\ ST_3]}\big(\llbracket [ST_3\ ST_3], \llbracket [ST_3\ ST_3], [ST_3\ T], \bullet \rrbracket\ \bullet\ \rrbracket\big) = \big(max(0.5, 0.5, 0.5), \text{ON}\big) = (0.5, \text{ON})$.

Step 5: $[ST_3\ ST_3]$ has been crossed for the second time (Fig. 19(a)). Once again, a time interval is counted to start decreasing the fluid level. In this case, $(\mu_E)^{max}_{[ST_3\ ST_3]}\big(\llbracket [ST_3\ ST_3], [ST_3\ T], \bullet \rrbracket\big) = \big(max(0.5, 0.5, 0.5),\ \text{ON}\big) = (0.5, \text{ON})$.

Next steps: After $[ST_3\ ST_3]$ has been crossed for the third time (Fig. 19(b)) and the arrow $[ST_3\ T]$ has been crossed, the arrow $[ST_3\ ST_3]$ becomes inactive (the counting of the time established for the drainage is finished). When crossing the arrow $[T\ \ V]$, the valve V is activated and blocks fluid from entering ($[START\ \ ST_1]$ has been inactive). In this case, $(\mu_E)^{max}_{[T\ V]}\big([START\ ST_1]\big) = \big(max(0.5, 0.3, 1), \text{ON}\big) = (1, \text{OFF})$.

The fluid level degrowth process will be omitted from this example, but it follows similarly. Other devices could be inserted in the system, such as accelerometers, alarms, among others.

6. Final remarks

Reversal Fuzzy Switch Graphs ($RFSG$s) are structures designed to model reactive systems which provide the activation and deactivation of resources. They are strongly based on the idea that the evolution of some systems results from the aggregation of previous information. This paper provides some adjustments in the current theory and some aggregation-based operations for them: Δ-union, Δ-intersection and extension. For Δ-union and Δ-intersection, some properties inherent to set theory were proven, however several others remain open: e.g. distributivity law and De Morgan's. An example of such properties is: *considering the union of two RFSGs, will a path composed of arrows active in one of them still remain active after the union?*

The most important operation presented here is *Extension*. Extensions of $RFSG$s models a way to increment the functionality of a given reactive system. It requires a wider investigation, for example: *how is the relationship between the extended graph with the original one?* or *what is the relation of extending reactive systems with aggregations?*

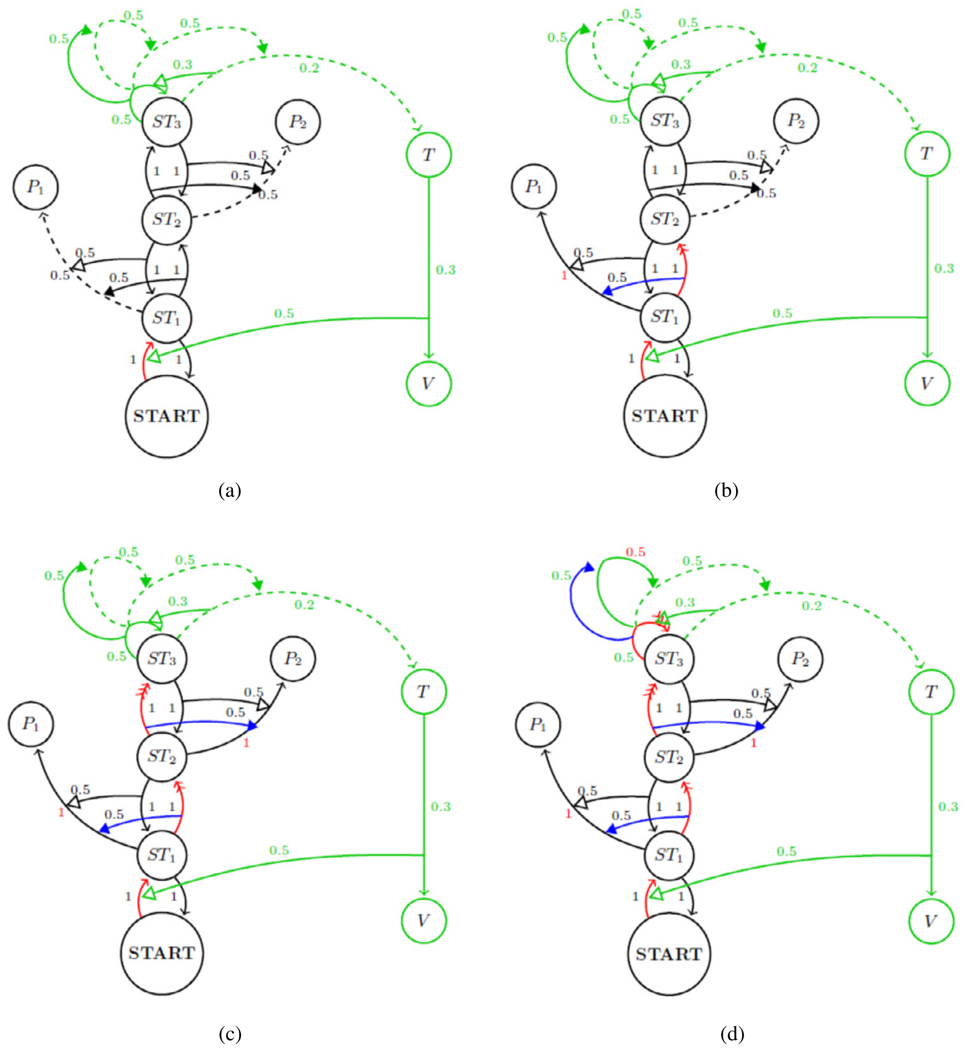


Fig. 18. Step by step of the fluid increasing and control system action.

We do not consider two parallel high-order arrow pointing to the same target arrow, such that one is connecting and the other is disconnecting. However, it is possible to establish state a priority order, like "disconnecting arrows act first", this is done in [10]. This idea is under development.

Since $RFRG$s are mathematical models for systems, some important concepts like: homomorphisms, simulation and bisimulation are under development.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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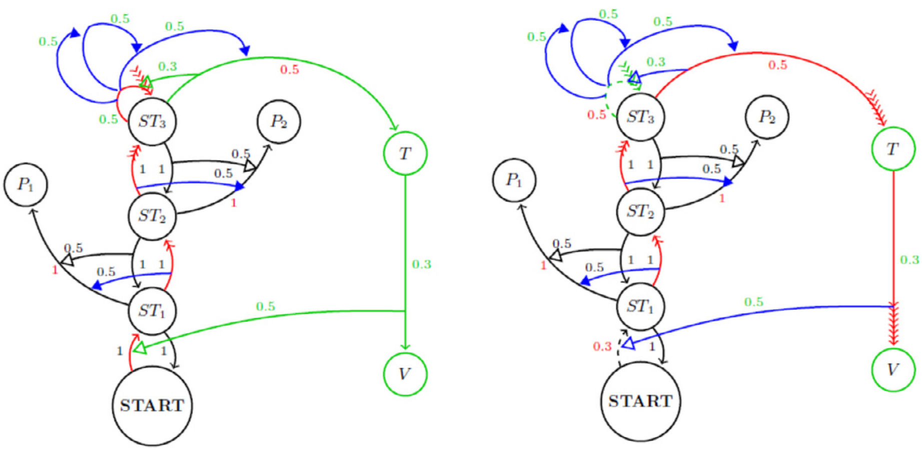

(a) (b)

Fig. 19. Step by step of the fluid increasing and control system action.

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